

Hugo Latourelle-Vigeant

Department of Statistics and Data Science – Yale University
New Haven, CT, USA

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Education

Ph.D. Statistics and Data Science

2024–Present (Expected 2030)

Yale University, New Haven, CT

- Advised by Professor Theodor Misiakiewicz

M.Sc. Mathematics and Statistics

2022–2024

McGill University, Montreal, QC

- Thesis topic: “The matrix Dyson equation for machine learning: Correlated linearizations and the test error in random features regression”
- Co-advised by Professor Courtney Paquette and Professor Elliot Paquette

B.Sc. Joint Honours Mathematics and Computer Science

2018–2022

McGill University, Montreal, QC

Research Interests

Random matrix theory, machine learning theory, statistical-computational trade-offs, and high-dimensional probability.

Publications & Preprints

- [1] **Latourelle-Vigeant, H.**, & Misiakiewicz, T. (2026). “Statistical-Computational Trade-offs in Learning Multi-Index Models via Harmonic Analysis.” *arXiv preprint arXiv:2602.09959*. (Submitted).
- [2] **Latourelle-Vigeant, H.**, & Paquette, E. (2026). “Dyson equation for correlated linearizations and test error of random features regression.” *Random Matrices: Theory and Applications*, 15(01), 2550026.

Talks & Presentations

- “Generalization, Memorization, and Overfitting in Kernel Denoising Score Matching in High Dimensions,” poster presentation, *Princeton Machine Learning Theory Summer School*, Princeton University, Princeton, NJ, August 2026.
- “Introduction to the Matrix Dyson Equation,” ETH Zurich, March 2024.
- “Matrix Dyson Equation for Correlated Linearizations,” *The many facets of random matrix theory* workshop, 2023 Canadian Mathematical Society Winter Meeting, Montreal, QC, December 2023.
- “Matrix Dyson Equation for Correlated Linearizations,” *Seminar in Machine Learning, Optimization and Random Matrix Theory*, McGill University, Montreal, QC, September 2023.
- “GD and Large Linear Regression: Concentration and Asymptotics for a Spiked Model,” 4th Undergraduate Student Research Conference, McGill University, Montreal, QC, August

2021.

Teaching Experience

Teaching Fellow for S&DS4100/6100 – Statistical Inference	Fall 2026
<i>Department of Statistics and Data Science, Yale University</i>	
Teaching Fellow for S&DS2420 – Theory of Statistics	Spring 2026
<i>Department of Statistics and Data Science, Yale University</i>	
Head Teaching Fellow for S&DS2410 – Probability Theory	Fall 2025
<i>Department of Statistics and Data Science, Yale University</i>	
Teaching Assistant for MATH 141 – Calculus 2	F2022, W2023, W2024
<i>Department of Mathematics and Statistics, McGill University</i>	
Graduate course assistant for MATH 463/563 – Convex Optimization	W2023, W2024
<i>Department of Mathematics and Statistics, McGill University</i>	
Graduate course assistant for MATH 417/517 – Linear Optimization	Fall 2022
<i>Department of Mathematics and Statistics, McGill University</i>	
Undergraduate course assistant for MATH 560 – Optimization	Winter 2022
<i>Department of Mathematics and Statistics, McGill University</i>	

*Note: W = Winter, S = Spring, F = Fall

Academic Service

Organization.....

Co-organizer for the Montreal RMT-OPT-ML Seminar	Fall 2023
<i>McGill University</i>	

Review.....

Main Track Reviewer	2026
<i>Conference on Neural Information Processing Systems (NeurIPS)</i>	
High-dimensional Learning Dynamics (HiLD) Workshop	2023, 2024, 2025, 2026
<i>International Conference on Machine Learning (ICML)</i>	
OPT Workshop on Optimization for Machine Learning	2022, 2023, 2024, 2025, 2026
<i>Conference on Neural Information Processing Systems (NeurIPS)</i>	

Volunteering.....

OPT Workshop on Optimization for Machine Learning	2020, 2021
<i>Conference on Neural Information Processing Systems (NeurIPS)</i>	

Work Experience

Data Science Intern	Summer 2024
<i>Caisse de dépôt et placement du Québec (CDPQ), Montreal, QC</i>	
○ Part of the natural language processing (NLP) team	
○ Conducted topic modeling and sentiment analysis on large-scale financial documents	

Awards and Scholarships

First-Class Honours in Mathematics and Computer Science <i>McGill University</i>	2022
Undergraduate Student Research Award (USRA) & FRQNT Supplement <i>NSERC & FRQNT</i>	2021
Major Entrance Scholarship in Science <i>Hydro-Québec</i>	2018